



Unit 3 · Outcome 2 · Sustainable Development

The beneficial and harmful impacts of a case study on Earth's four interrelated systems

KK07 VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. Earth systems thinking Beneficial & harmful 5 practice questions · 20 marks ♦ The exact dot point

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



01 The four systems, defined

Environmental scientists model Earth as four interrelated systems. Every environmental impact lands in one or more of them. Know the one-line definition of each cold — examiners give easy marks for it, and you can't trace an impact if you can't name the system it lands in.

The layer of gases surrounding Earth — and everything carried in it: oxygen, carbon dioxide, methane, water vapour, dust and heat. Think weather, greenhouse gases, air quality and microclimate.

All the regions of Earth where life exists — every plant, animal, fungus and microbe, on land, in water and in the soil. If it's alive (or was), it's biosphere.

All the water on Earth in liquid or solid form — rivers, wetlands, lakes, oceans, groundwater and ice. (Water vapour belongs to the atmosphere — a classic trap.)

The solid outer Earth — rock, sediment, soil and landforms. Erosion, salinity in the soil, sediment and the riverbanks themselves all live here.

Below is the whole Boorla catchment as a single Earth-system stack. Drag to orbit it — sky, life, water and land, the four systems sitting one on top of another.

⚠ WATCH OUT

Analogy — the human body VCAA leans on this comparison, so use it. Earth's four systems are like the body's organ systems — circulatory, respiratory, digestive, nervous. They look separate, but they're plumbed together. Sprint up a flight of stairs and your muscles (one system) force your heart and lungs (two more) to respond within seconds. You can't change one without the others reacting. Earth is the same: that's what "interrelated" means.

02 The big idea: systems are interrelated

This is the heart of the dot point — and the place most marks are won or lost. The four systems are not four separate boxes. Matter and energy move between them constantly, so a change in one cascades into the others. VCAA calls this Earth systems thinking, and the command verbs ("analyse", "evaluate") demand you show the links, not just list impacts in silos.

◆ NOTE

What "interrelated" buys you in the exam A weak answer says: "Watering the wetland helps the plants." One system, no link, ~1 mark. A strong answer traces the ripple: "Water enters the hydrosphere → re-wets the soil of the lithosphere → river red gums in the biosphere recover → their growth draws down CO₂ from the atmosphere." Four systems, three links — that's where the marks live.

The cascade below makes it physical. Press Send environmental flow and watch one action move through all four systems. Then flip the toggle to a blackwater event — the same water, delivered badly, running the cascade in reverse and turning beneficial into harmful.

◆ NOTE

Healthy flow — the beneficial cascade Delivered in the cool months as a gentle pulse, the water re-wets the floodplain slowly. Red gums and reeds green up, fish and frogs breed, the wetland sediments lock away carbon. The link runs hydrosphere → lithosphere → biosphere → atmosphere, and it's good news at every step.

WATER

03 The impacts ledger — beneficial & harmful

For the exam you need a stocked toolkit: at least one beneficial and one harmful impact for each system, with the mechanism (the "because..."). Here is the Boorla ledger. Notice the recurring theme — the same restoration is a gift to some systems and a cost to others, and even within one system it can be good long-term but bad short-term.

⚠ WATCH OUT

Analogy — the household budget An environmental flow is a finite payment. Spend it on the wetland and the wetland's "account" (biosphere, lithosphere) grows — but the town and farm accounts shrink. Sustainability isn't pretending there's no cost; it's showing you've weighed the credit against the debit across every account. That balancing act is the higher-mark skill.

05 Practice: sort the impact

★ PRACTICE & SELF-MARK

The single most-tested micro-skill here: take an impact statement and (1) name the system it lands in, then (2) call it beneficial or harmful. Work through all eight. Instant feedback; your score feeds the dock.

06 Decode the command word

◆ COMMAND WORDS

The verb tells you how hard to work. "Identify the system" wants one word; "evaluate the impacts" wants a weighed judgement across systems. Tap a verb to see what it demands for this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

07 Build an answer: the P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

For an "impacts on Earth's systems" answer, the reliable structure is P·E·L: Point (name the system + beneficial/harmful), Example (the specific Boorla mechanism), Link (cascade it to another system). Draft each part, then reveal the model. Prompt: Describe one harmful impact of the Boorla environmental flow on the atmosphere, and link it to another system. (3 marks) P The environmental flow has a harmful impact on the atmosphere, as it increases greenhouse-gas emissions. E When the released water saturates the wetland's organic-rich sediments, those sediments become anaerobic (oxygen-poor), and microbes there break down organic matter to release methane (CH₄) — a greenhouse gas far more potent than CO₂. L This impact originates in the hydrosphere (the added water) acting on the lithosphere (the saturated sediment), showing how an atmospheric impact is driven by changes cascading from two other interrelated systems.

08 Six exam traps

EXAM TRAPS

"It helps the plants and the water." Which sphere? Always pin each impact to atmosphere / biosphere / hydrosphere / lithosphere by name. If the question says "beneficial and harmful", a one-sided answer caps your marks. Give both, even when one feels obvious. The dot point says interrelated. Higher marks require you to link at least two systems (the cascade), not list four disconnected facts. Liquid/ice water = hydrosphere. Water vapour and humidity = atmosphere. Evaporation is literally water moving between the two. Fish live in water and worms live in soil, but they are biosphere. The system is defined by what the thing is, not where it sits. "It's good for the atmosphere" earns little. "... because re-grown red gums photosynthesise and sequester CO₂" earns the mark. State the cause.

09 One-screen cheat sheet

RECAP / CHEAT SHEET

+ wetland refills, salinity diluted, water table recharged · – less water for town/farms, blackwater drops O₂ + red gums, waterbirds, native fish & microbes return · – fish kills (hypoxia), carp & mozzies benefit + carbon sink, cooler/humid microclimate, less dust · – methane from anaerobic sediment, CO₂ pulse + less erosion, soil rebuilds, dryland salinity eased · – acid sulfate soils, bank erosion, salt mobilised

NOTE

The one move that lifts your band Always cascade: water (hydro) → soil (litho) → plants (bio) → carbon (atmos). Name the system, give the mechanism, link to a neighbour.

10 Exam practice · weak vs strong

Five VCAA-style questions, 20 marks total. For each, try it yourself first, then reveal the weak response, the annotated strong response and the marking guide. Mark your own attempt — the dock keeps your running total.

U3·02 · KK07 SUSTAINABLE DEVELOPMENT

The beneficial and harmful impacts of a case study on Earth's four interrelated systems

Pull one thread of an ecosystem and the whole web moves. This dot point asks you to take your case study — here, the Boorla Wetlands recovery — and trace how its impacts spread, for better and for worse, through the atmosphere, biosphere, hydrosphere and lithosphere.

◆ NOTE

The exact dot point "The beneficial and harmful impacts of the selected case study on Earth's four interrelated systems — atmosphere, biosphere, hydrosphere and lithosphere."

◆ NOTE

Our running case study — recap from KK01 The Boorla Wetlands sit on the floodplain of the regulated Yarrin River, beside the town of Connawarra. Decades of dam storage and irrigation diversions starved the wetland of its natural floods; it shrank, salted up and lost its waterbirds. The management response: return an environmental flow — water deliberately released to re-flood the wetland — alongside revegetation and a restored wetting-and-drying cycle. This module asks the Earth-systems question: when that water arrives, what does it do to each of the four systems — and is it good or bad?

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.